Jiazheng Zhu

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Research Profile

I am a Ph.D. candidate in astrophysics at USTC, studying **time-domain astrophysics and optical transients**, with a particular focus on **tidal disruption events (TDEs)**. My research combines wide-field transient surveys, photometric and spectroscopic follow-up, and multiwavelength analysis to connect transient discovery with physical interpretation.

My future research aims to use TDEs to probe black holes across new mass and redshift regimes, particularly **intermediate-mass black holes (IMBHs)** and **high-redshift TDEs**. By combining deep surveys, including the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), with rapid follow-up, I aim to uncover faint and poorly sampled transient populations and investigate black-hole demographics and transient accretion physics.

Education

University of Science and Technology of China 2021–2027.6 (expected)

Ph.D. Candidate in Astrophysics, School of Physical Sciences; Hefei, China

Integrated master's–doctoral program. Advisors: Prof. Tinggui Wang and Prof. Ning Jiang.

University of Science and Technology of China 2017–2021

B.S. in Astronomy, School of Physical Sciences; Hefei, China

Research Interests

- **Time-domain astrophysics and optical transients:** discovery and physical characterization through wide-field surveys and multiwavelength follow-up.
- **Tidal disruption events:** observational diversity, transient accretion, host-galaxy environments, and probes of otherwise quiescent black holes.
- **IMBHs and high-redshift TDEs:** extending black-hole searches to new mass, host-galaxy, and redshift regimes.
- **Faint and unexplored transient populations:** faint nuclear transients and unusual accretion events accessible to increasingly deep surveys.

Honors & Awards

Doctoral Fellowship 2025, 2026

National Advanced Talent Cultivation Center for Physics, USTC

National Scholarship for Graduate Students 2023

Ministry of Education of the People's Republic of China

Outstanding Volunteer 2023

USTC Summer Camp for Outstanding Undergraduates

Publications

Four first-author publications and 20 co-authored entries, including one correction. Paper titles link to their DOI records.

First-author Publications

[1] **Zhu, J.**, Xu, Z., Jiang, N., et al. (2026). **Discovery of a Featureless Tidal Disruption Event at $z \sim 1$ with the Wide Field Survey Telescope**. *The Astrophysical Journal Letters*, 1004, L10.

[2] **Zhu, J.**, Jiang, N., Wang, Y., et al. (2025). **Ultraviolet Spectral Evidence for Ansky as a Slowly Evolving Featureless Tidal Disruption Event with Quasiperiodic Eruptions**. *The Astrophysical Journal Letters*, 994, L16.

[3] **Zhu, J.**, Jiang, N., Wang, T., et al. (2023). **AT 2023clx: The Faintest and Closest Optical Tidal Disruption Event Discovered in Nearby Star-forming Galaxy NGC 3799**. *The Astrophysical Journal Letters*, 952, L35.

[4] **Zhu, J.**, Jiang, N., Dong, S., et al. (2023). **SN 2017egm: A Helium-rich Superluminous Supernova with Multiple Bumps in the Light Curves**. *The Astrophysical Journal*, 949, 23.

Co-author Publications

[5] Huang, S., Wang, T., Jiang, N., Wang, Y., Sheng, Z., Xia, T.-Y., **Zhu, J.**, et al. (2026). **A Unified Timescale Relation for Quasiperiodic Eruptions and Repeated Nuclear Transients**. *The Astrophysical Journal Letters*, 1009, L23.

[6] Wang, Y., Wang, T., Huang, S., **Zhu, J.**, Jiang, N., et al. (2026). **A Star's Death by a Thousand Cuts: The Runaway Periodic Eruptions of AT2023uqm**. *Science Bulletin*, 71, 4645–4656.

[7] Liu, H., Sun, L., Jiang, N., Shu, X., Wang, Y., Wang, T., Cutri, R. M., Dou, L., Zhang, F., **Zhu, J.**, et al. (2026). **An Obscured Tidal Disruption Event Uncovered by Its Mid- and Near-Infrared Dust Echo in a Star-forming Galaxy**. *The Astrophysical Journal*, 1005, 178.

[8] Li, D., Zhang, W., Yang, J., Chen, J.-H., Yuan, W., Cheng, H., Xu, F., Shu, X., Shen, R.-F., Jiang, N., **Zhu, J.**, et al. (2026). **A Fast Powerful X-ray Transient from Possible Tidal Disruption of a White Dwarf**. *Science Bulletin*, 71, 538–546.

[9] Wang, J., Huang, M., Xue, Y., Jiang, N., Huang, S., Wang, Y., **Zhu, J.**, et al. (2026). **A Tidal Disruption Event from an Intermediate-mass Black Hole Revealed by Comprehensive Multi-wavelength Observations**. *Nature Communications*, 17, 2007.

[10] Wu, M., Jiang, N., **Zhu, J.**, Luo, D., Dou, L., Wang, T. (2025). **A Torus Remnant Revealed by the Infrared Echo of Tidal Disruption Event AT 2019qiz: Implications for the Missing Energy and Quasiperiodic Eruption Formation**. *The Astrophysical Journal Letters*, 988, L77.

[11] Huang, S., Wang, T., Jiang, N., Shen, R.-F., Chen, Z., Wang, Y., **Zhu, J.**, et al. (2025). **Unveiling the Cosmic Dance of Repeated Nuclear Transient ASASSN-14ko: Insights from Multiwavelength Observations**. *The Astrophysical Journal*, 988, 237.

[12] Lin, J., Wang, T., Cai, M., et al. (2025). **Minute-cadence Observations of the Galactic Plane with the Wide Field Survey Telescope (WFST): Overview, Methodology, and Early Results**. *The Astrophysical Journal Supplement Series*, 278, 29.

[13] Jiang, N., Luo, D., **Zhu, J.**, Cutri, R. M. (2025). **The Extraordinary Long-lasting Infrared Echo of PS16dtm Reveals an Extremely Energetic Nuclear Outburst**. *The Astrophysical Journal Letters*, 980, L17.

[14] Liu, Y., Fan, L., Hu, L., Lu, J., Lu, Y., Xu, Z., **Zhu, J.**, et al. (2025). **Classification of Real and Bogus Transients Using Active Learning and Semi-supervised Learning**. *Astronomy & Astrophysics*, 693, A105.

Co-author Publications (continued)

[15] Jin, C.-C., Li, D.-Y., Jiang, N., Dai, L.-X., Cheng, H.-Q., **Zhu, J.**, et al. (2025). **An Intermediate-mass Black Hole Lurking in a Galactic Halo Caught Alive during Outburst**. *arXiv e-prints*, arXiv:2501.09580.

[16] Sun, L., Jiang, N., Dou, L., Shu, X., **Zhu, J.**, et al. (2024). **Recurring Tidal Disruption Events a Decade Apart in IRAS F01004-2237**. *Astronomy & Astrophysics*, 692, A262.

[17] Lin, Z., Jiang, N., Wang, T., Kong, X., Li, D., He, H., **Zhu, J.**, et al. (2024). **The Unluckiest Star: A Spectroscopically Confirmed Repeated Partial Tidal Disruption Event AT 2022dbl**. *The Astrophysical Journal Letters*, 971, L26.

[18] Wang, Y., Wang, T., Jiang, N., Zhang, X., **Zhu, J.**, et al. (2024). **ASASSN-18ap: A Dusty Tidal Disruption Event Candidate with an Early Bump in the Light Curve**. *The Astrophysical Journal*, 966, 136.

[19] Huang, S., Jiang, N., Lin, Z., **Zhu, J.**, Wang, T. (2024). **Correction to: AT2018dyk Revisited: A Tidal Disruption Event Candidate with Prominent Infrared Echo and Delayed X-ray Emission in a LINER Galaxy**. *Monthly Notices of the Royal Astronomical Society*, 531, 4793.

[20] Huang, S., Jiang, N., **Zhu, J.**, Wang, Y., Wang, T., et al. (2024). **AT 2023lli: A Tidal Disruption Event with Prominent Optical Early Bump and Delayed Episodic X-Ray Emission**. *The Astrophysical Journal Letters*, 964, L22.

[21] Jiang, N., Zhou, Z., **Zhu, J.**, Wang, Y., Wang, T. (2023). **Two Candidate Obscured Tidal Disruption Events Coincident with High-energy Neutrinos**. *The Astrophysical Journal Letters*, 953, L12.

[22] Lin, Z., Jiang, N., Kong, X., Huang, S., Lin, Z., **Zhu, J.**, et al. (2022). **The Luminosity Function of Tidal Disruption Flares for the ZTF-I Survey**. *The Astrophysical Journal Letters*, 939, L33.

[23] Dou, L., Jiang, N., Wang, T., Shu, X., Yang, H., Pan, Z., **Zhu, J.**, et al. (2022). **X-ray View of a Merging Supermassive Black Hole Binary Candidate SDSS J1430+2303: Results from the First ∼200 Days of Observations**. *Astronomy & Astrophysics*, 665, L3.

[24] Wang, Y., Jiang, N., Wang, T., **Zhu, J.**, et al. (2022). **Discovery of ATLAS17jrp as an Optical-, X-Ray-, and Infrared-bright Tidal Disruption Event in a Star-forming Galaxy**. *The Astrophysical Journal Letters*, 930, L4.

Conference Presentations

Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era
Heraklion, Crete, Greece 2–6 September 2024

- **Oral presentation:** "Mid-Infrared Outbursts in Nearby Galaxies (MIRONG)."
- **Poster:** AT 2023clx as a representative of the faint TDE population.

Observational Experience & Telescope Time

Observing Experience

- **Wide Field Survey Telescope (WFST), 2023–Present.** One week of on-site observing and six weeks of remote observing in total. Experience includes observing assistance, monitoring observing conditions, survey operations, and real-time image-quality assessment.
- **Lijiang 2.4-m Telescope, 2024.** Two nights of on-site observing.

Principal Investigator Programs

- **Gemini North, 2024B** `GN-2024B-DD-105`
  "Towards the first optical discovery of an IMBH-TDE at its Eddington accretion phase in a dwarf starburst galaxy."
  Approved time: **2 hours**.
- **Swift UVOT/XRT, 2023–2026.** Target-of-opportunity requests for individual transients. Total approved time: **129 ks**.

Co-Investigator Programs

- **JWST** `JWST-DD-12569`
  "Are LFBOTs Dusty? Constraining the Nature of IR Emission in the Luminous Fast Blue Optical Transient AT2026dbl."
  Approved time: **5.2 hours**.
- **Gemini North, 2024A** `GN-2024A-DD-103`
  "Towards a Deep Spectroscopic View of the First X-Ray IMBH-TDE Candidate Discovered at Early Phase."
  PI: Ning Jiang. Approved time: **3 hours**.
- **Gemini North, 2023A** `GN-2023A-Q-110`
  "Spectroscopic Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **4.5 hours**.
- **Gemini North, 2022A** `GN-2022A-DD-101`
  "Spectroscopic Diagnosis of Optical flares from a merging supermassive black hole binary."
  PI: Zhen Pan. Approved time: **2 hours**.
- **HST** `HST-GO/DD-18213`
  "Imaging of EP241021a during the flattening in the optical light curve: an exotic gamma-ray burst or a peculiar jetted tidal disruption event?"
  Approved time: **2 orbits**.
- **HST** `HST-GO/DD-17933`
  "Turn-on AGN or IMBH-TDE? UV Spectroscopic Observations May Tell."
  Approved time: **4 orbits**.
- **HST** `HST-GO/DD-17849`
  "Probing the UV spectral evolution of a unique TDE with early bright X-ray emission."
  Approved time: **4 orbits**.
- **HST** `HST-GO/DD-17609`
  "A UV spectroscopic view of the 2024 outburst of AT2020afhd: A Unique Opportunity to Test the Reprocessing Model of Tidal Disruption Event."
  Approved time: **2 orbits**.
- **VLA, 2026A** `VLA/26A-638`
  "VLA Observation of a Unique Off-nuclear TDE Candidate."
  Director's Discretionary Time; proposal date: 2 June 2026. Approved time: **2 hours at Priority B**, plus **8 hours at Priority C** conditional on detection in the initial observation.

- **LCOGT, 2023A** `NAOC2023A-005`
  "Continuous Daily Optical Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **35 hours**.
- **LCOGT, 2022B** `NAOC2022B-002`
  "Further Daily Optical Monitoring of an unprecedented SMBH Binary at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **20 hours**.
- **MMT/Binospec, 2025B** `UAO-S111-25B`
  "Spectroscopic Identification of Candidate Tidal Disruption Events Caused by Intermediate-Mass Black Holes."
  PI: Ann Zabludoff. Approved time: **1.5 nights** of queue observing.

Additional collaborative observing proposals include P200, Swift, XMM-Newton, and the Lijiang 2.4-m telescope; program details are not listed here.

Research & Technical Skills

- **Surveys and transient discovery:** transient detection and classification, survey data mining, candidate selection, light-curve analysis, variability characterization, and multiband photometry.
- **Imaging and photometry:** CCD calibration, image stacking, astrometry, image subtraction, difference imaging, and photometric measurements.
- **Spectroscopy:** optical/UV spectral analysis, continuum and emission-line characterization, and transient classification.
- **SED and physical modeling:** multiwavelength spectral energy distribution fitting, blackbody and continuum modeling, Markov chain Monte Carlo (MCMC), and Bayesian inference.
- **Host galaxies:** galaxy photometry and morphology, stellar-population analysis, host-galaxy SED fitting, and spectral fitting.
- **Scientific computing:** astronomical data analysis, scientific computing, and visualization, primarily in Python.

Teaching Experience

Teaching Assistant, Mechanics A 2023
School of Physical Sciences, University of Science and Technology of China

Teaching Assistant, Thermal Physics A 2021
School of Physical Sciences, University of Science and Technology of China

Service & Outreach

Astronomy Summer Camp Volunteer 2023, 2026
University of Science and Technology of China

Astronomy Association Member
Wenzhou, China (my hometown)

Campus Service & Astronomy Outreach
University of Science and Technology of China

Actively participate in university and astronomy department student activities and promote astronomy through science communication and outreach.